MOHAMMED MAHYOUB

Senior ITS & Systems Integration Engineer | Applied AI & Computer Vision

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PROFESSIONAL SUMMARY

Senior Service Engineer with 14 years of multidisciplinary engineering experience, including over 7 years at VITRONIC Machine Vision supporting intelligent transportation, smart infrastructure and traffic enforcement systems across the UAE. I integrate field devices, networks and backend platforms, and lead commissioning, calibration and structured root-cause analysis. Currently pursuing an MSc in Artificial Intelligence at UWE Bristol, focusing on computer vision, data quality, leakage-controlled evaluation and efficient neural networks for smart mobility.

CORE EXPERTISE

ITS & Traffic Enforcement | Systems Integration | FAT / SAT, Commissioning & Calibration | Machine Vision & LiDAR | Root-Cause Analysis | Networking (Cisco, MikroTik, MPLS, VoIP/IMS) | Virtualisation & Data Centre (VMware ESXi, iSCSI) | Embedded Systems & Automation | Python, PyTorch & Machine Learning

PROFESSIONAL EXPERIENCE

Senior Service Engineer - Smart Infrastructure & ITS Systems | VITRONIC Machine Vision

Oct 2019 - Present | United Arab Emirates

- Support an installed base of 800+ intelligent transportation and traffic enforcement systems across the UAE.
- Integrate cameras, LiDAR, sensors, control units, networks, databases, and backend platforms into reliable end-to-end systems.
- Execute FAT, SAT, commissioning, calibration, interface validation, acceptance testing, and performance verification.
- Diagnose complex hardware, software, communication, database, and system-interface faults using structured root-cause analysis.
- Coordinate incident response, preventive maintenance, upgrades, and configuration changes with customers, vendors, and engineering teams.
- Prepare technical reports, test records, incident analyses, and system handover documentation.

Electronics, Embedded Systems & Automation Engineer | Electronic Science Company

Jun 2015 - Aug 2019 | Al Hudaydah, Yemen

- Designed, assembled, tested, and troubleshot electronic circuits, embedded control systems, and sensor-based automation.
- Programmed PIC, AVR, and Arduino microcontrollers; integrated sensors, displays, communication modules, controllers, and PLCs.
- Performed PCB prototyping, hardware debugging, functional testing, and technical validation.
- Developed VB.NET desktop applications for monitoring, diagnostics, reporting, and workflow automation, with deployment and customer support.

Independent Engineering Consultant - Network Infrastructure & Technical Projects | Independent Engineering Projects

Mar 2012 - May 2018 | Part-time / Project-based

- Designed and supported wired and wireless infrastructure using routing, switching, firewall, VPN, MikroTik, and Ubiquiti, including coverage and point-to-point link planning.
- Designed, built, and validated MPLS, OSPF, BGP, VRF, and traffic-engineering architectures using Cisco platforms and GNS3.
- Designed and deployed photovoltaic power systems with battery storage and inverters, including solar supply for remote telecommunications sites.
- Developed engineering tools for RF calculations and reporting; prepared network diagrams, configurations, and technical documentation.

ENGINEERING PROJECTS

Independent implementations and research; linked titles open the complete case studies.

ITS Systems Integration & Traffic Enforcement Support

Solar Energy Systems - Design, Sizing & Deployment

Efficient Neural Network Architectures for Vehicle Type Classification

Industrial Automation & PLC-Based Control Systems

Assessment code private

Vehicle Type Classification Using NGSIM US-101

Engineering Operations Suite - Monitoring, Diagnostics & Service Management

Implemented apps; shared integration proposed

Reconnaissance Robot - RGB-D Mapping & Remote Control

Wireless Link Planning & RF Engineering

Historical implementation; later RF review

Autonomous Quadcopter UAV - Flight Control, Telemetry & FPV

Enterprise Wireless & Network Infrastructure Architecture

Team implementation

Software-Defined Networking - Automation & Video Streaming

PIC-Based Equipment Room Monitoring & Control System

Includes PIC peripherals and EEPROM storage

IMS Multimedia Services - VoIP, IPTV & VoLTE

Network Infrastructure Design - MPLS Backbone

Team implementation

Software-Defined Data Center (SDDC) - Design & Implementation

RFID Library Automation System (ISO 15693)

Team implementation

NGSIM archived publication: DOI 10.5281/zenodo.23033235 | USV Systems Integration - separate study notes

EDUCATION

MSc Artificial Intelligence | University of the West of England (UWE Bristol)

May 2026 - Jun 2028 | In progress - machine learning, neural networks, computer vision, model evaluation and efficiency

BEng Electrical Engineering - Communications & Electronics | Sana'a University

Sep 2012 - Jun 2017 | UAE Ministry of Education degree equivalency obtained

CERTIFICATIONS & RECOGNITION

- Certificate of Appreciation - Dubai Police HQ, UAE (2023)

Previously earned: CCNA Routing & Switching (2019; valid until 2022) | CompTIA A+ (2019; valid until 2022)

- PIC Microcontroller Programming - mikroC & Proteus (training)

TECHNOLOGIES & LANGUAGES

Technologies: Python, PyTorch, scikit-learn, Pandas, VB.NET, mikroC, PIC/AVR/Arduino, PLC, ArduPilot/MAVLink, Cisco, MikroTik, Ubiquiti, GNS3, OPNET, Cisco CUCM/CME, VMware ESXi, FreeNAS/iSCSI, Citrix XenApp, Exchange, Active Directory.
Languages: Arabic (Native / Bilingual) | English (Professional Working Proficiency)